WRITING AND BALANCING EQUATIONS

These show the chemical change that occurs in a reaction.

Rules

1. ***Write the equations in words to determine the reactants and products***.

   - This step may be omitted.

2. ***Write the chemical equation using the correct chemical formulas.***

3. ***Subscripts are put after each compound to show the physical state the compound is in***.

   (s) = solid **(l) = liquid** **(g) = gas** **(aq) = in solution**

   e.g. $H_2O_{(g)}$ = steam $H_2O_{(s)}$ = ice $NaCl_{(aq)}$ = salt dissolved in solution.

4. ***Count the number and types of atoms on both sides of the equation***.

   - Polyatomic ions (like SO_4^{2-}), which do not change from the reactants to the products side, should be considered as a group.

5. ***Add numbers in front of the formulas***.

   - Make sure that the total number of atoms of reactants is the same as the total number of atoms of products.

   - Tick the elements off as you balance them.

   - Begin with the element, other than hydrogen or oxygen, that has the greatest number of atoms in any reactant or product.

   - Make sure you never change the formula of a chemical substance to balance an equation.

6. ***Do a final check.***

Examples

Balance the following equations.

1. Solid zinc hydroxide + sulphuric acid solution → zinc sulphate solution + water

2. Lead (II) nitrate solution + potassium phosphate solution → solid lead (II) phosphate + potassium nitrate solution

3. heptane (C_7H_{16}) + oxygen gas → carbon dioxide gas + water vapour

4. Copper metal + nitric acid solution → copper (II) nitrate solution + nitrogen dioxide gas + water